



Results of combined photodynamic therapy (PDT) and high dose rate brachytherapy (HDR) in treatment of obstructive endobronchial non-small cell lung cancer (NSCLC)

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KEYWORDS

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Summary

Introduction: We reviewed the outcome of combined photodynamic therapy (PDT) and high dose rate brachytherapy (HDR) for patients with symptomatic obstruction from endobronchial non-small cell lung cancer.

Methods: Nine patients who received combined PDT and HDR for endobronchial cancers were identified and their charts reviewed. The patients were eight males and one female aged 52–73 at diagnosis, initially presenting with various stages of disease: stage IA ($N=1$), stage IIA ($N=1$), stage III ($N=6$), and stage IV ($N=1$). Intervention was with HDR (500 cGy to 5 mm once weekly for 3 weeks) and PDT (2 mg/kg Photofrin®, followed by 200 J/cm² illumination 48 h post-infusion). Treatment group 1 (TG-1, $N=7$) received HDR first; Treatment group 2 (TG-2, $N=2$) received PDT first. Patients were followed by regular bronchoscopies.

Results: Treatments were well tolerated, all patients completed therapy, and none were lost to follow-up. In TG-1, local tumor control was achieved in six of seven patients for: 3 months (until death), 15 months, 2+ years (until death), 2+ years (ongoing), and 5+ years (ongoing, $N=2$). In TG-2, local control was achieved in only one patient, for 84 days. Morbidities included: soft-tissue contraction and/or other reversible benign local tissue reactions ($N=8$) and photosensitivity reactions ($N=2$).

Conclusions: Combined HDR/PDT treatment for endobronchial tumors is well tolerated and can achieve prolonged local control with acceptable morbidity when PDT follows HDR and when the spacing between treatments is 1 month or less. This treatment regimen should be studied in a larger patient population.

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Introduction

Most patients with non-small cell lung cancer (NSCLC) are diagnosed in advanced stage [1], when standard of care

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treatment generally consisting of chemotherapy and external beam radiation therapy (EBRT) cannot readily control this aggressive disease [2]. Local recurrence after first-line treatment often presents as endobronchial tumor, which is highly symptomatic due to its occlusion of the airways, leading to significant morbidity and mortality if untreated. Often, this airway compromise is accompanied by potentially life-threatening hemoptysis [3]. Primary early-stage NSCLC can also present as an obstructive mass with occlusive symptoms and hemoptysis, and may be inoperable due to its location or other factors. Endobronchial tumor growth adds further burden to a patient population in which lung function has often already been compromised by primary tumor and pre-existing chronic lung disease.

Measures taken to improve pulmonary function and eliminate hemoptysis can improve quality of life. In primary endobronchial tumors, the goal is to cure, while in recurrent cases, further treatment is often considered palliative and may not significantly improve patient survival. Possible interventions for obstructive endobronchial lesions include: EBRT, where meaningful doses are difficult to deliver in recurrent cases because prior courses have reached dose limits; chemotherapy, which is unlikely to succeed in a tumor in which first- and second-line agents have already failed and which is unnecessary for small early-stage primary tumors; and specifically designed bronchoscopic interventions, which have been more successful in this clinical situation.

Available bronchoscopic procedures include both ablative and non-ablative technologies, any of which can be used to mitigate airway obstruction. Bronchoscopic options include stenting, cryotherapy, electrocautery, argon plasma coagulation, Nd:YAG laser therapy, high dose rate brachytherapy (HDR), and photodynamic therapy (PDT) [4]. As there have been no large-scale randomized trials comparing these different modalities; equipment availability, user expertise, and the emergent nature of the obstruction often dictate the choice of intervention. Of the available therapies, stenting, electrocautery, argon plasma coagulation, and laser therapy offer the most immediate results, while the tissue effects of cryotherapy, HDR and PDT have a delayed onset [5]. However, both HDR and PDT appear to offer more prolonged local control of disease than the other bronchoscopic techniques [6,7]. In addition, the use of multiple bronchoscopic modalities has been shown to significantly increase survival compared to the use of only a single modality [8] and sequential PDT/HDR treatments have been used to effectively control the growth of limited endobronchial carcinoma [9]. We postulated that sequential use of HDR and PDT would maximize local tumor control and minimize the overall number of interventions required for treatment of endobronchial NSCLC; we now report the outcomes of this intentionally combined treatment in nine patients.

Materials and methods

Patients

We identified nine patients in the Leo Jenkins Cancer Center's computerized record system (from 1/2001 to 8/2008) who had received both photodynamic therapy (PDT) and

high dose rate brachytherapy (HDR) for treatment of endobronchial tumors. These patients' clinic and hospital records were thoroughly reviewed to determine the presentation and course of their disease and treatments, the details of their PDT and HDR interventions, and any adverse events or morbidities attributable to the PDT/HDR treatment regimen.

These nine patients were eight males and one female aged 52–73 at initial diagnosis who presented for PDT/HDR treatment with various stages of endobronchial NSCLC. All of these patients presented with occlusive symptoms with or without hemoptysis. They were treated in our clinic with full PDT and HDR treatments given 1–2 months apart, except for patients 1-7 and 2-2, who stopped their HDR treatments early (see Table 3). They were divided into two groups based on the sequence of their combined treatment. Patients in treatment group 1 (TG-1, $N=7$) received HDR before PDT, while those in treatment group 2 (TG-2, $N=2$) received PDT before HDR. One patient (patient 1-2) received HDR followed by PDT, with a second HDR treatment 2 months later. He was assigned to group 1 based on the order of his first two treatments. Table 1 displays the patients' initial diagnoses and the primary treatments they received, if any, before receiving HDR/PDT.

Treatments

High dose-rate brachytherapy (HDR)

HDR is a form of radiation therapy wherein a radioactive source (usually iridium-192) is placed in or near a tumor to deliver ionizing radiation and is withdrawn after a set period of time depending on the desired dose. For endobronchial tumors, the HDR source is loaded into the treatment area remotely through a catheter which has been bronchoscopically placed across the tumor, using diagnostic X-rays to confirm placement. The procedure is generally repeated, delivering the total dose in three fractions spaced 1 week apart. This is a minimally invasive procedure, which can be performed on an outpatient basis under light sedation [10].

In this study, HDR intervention consisted of flexible bronchoscopic introduction of a single HDR catheter. The patient was then immediately brought to the HDR unit, and 500 cGy was delivered to 0.5-cm depth via a Nucletron remote after-loading HDR unit. The tumor bed, defined at bronchoscopy, plus at least 1 cm proximally and distally was irradiated. Orthogonal diagnostic X-ray films were used for verification and dose calculation. Two additional HDR treatments were delivered at weekly intervals for a total dose of 15 Gy in three fractions.

Photodynamic therapy (PDT)

PDT utilizes monochromatic light, generally from a laser source, to locally excite a photosensitizing agent (photosensitizer, PS) which has been delivered systemically or topically to the tumor. Normal tissues tend to eliminate PS faster than tumor cells, so concentrations in malignant tissue are higher than in surrounding healthy tissue [11]. The light source activates the PS, releasing destructive superoxide and hydroxyl radicals which lead to apoptosis, necrosis, vascular occlusion, and activation of immune response [11,12]. Tumor destruction

Table 1 Patients at initial presentation.

Patient	Initial diagnosis (age)	Stage, histology, grade	Location	Initial treatments (before HDR/PDT)	Date of recurrence (age)
Treatment group 1 (TG-1, HDR before PDT)					
1-1	1994 (58), 2000 (64)	IIIB sq, mod	LUL, RUL	Surgical resection × 2 (LUL, RUL)	9/23/2003 (68)
1-2	12/3/2004 (69)	IV sq, poor	LUL, R Hilum, Pretracheal lymph node	Laser therapy, EBRT (RUL); LUL untreated per patient request	7/13/2006 (71)
1-3	5/17/2002 (73)	IIIB sq	LMSB, Node in mediastinum (5 cm × 4 cm)	Chemotherapy, EBRT	1/2/2003 (73)
1-4	6/2006 (64)	IIB/IA (dual primaries) sq, poor	RUL/LUL (only LUL treated by PDT/HDR)	Chemotherapy, EBRT	9/1/2006 (64)
1-5	3/2003 (63)	IIIB sq, mod	RUL	Chemotherapy, APC	2/9/2004 (63)
1-6	3/21/2003 (61)	IIA sq, undif	RLL, RBI	None	—
1-7	12/17/2002 (52)	IIIB sq	Distal Trachea, Proximal LMSB and RMSB	None	—
Treatment group 2 (TG-2, PDT before HDR)					
2-1	9/2005 (55)	IIIB sq, poor	LUL, LLL, Lingula, LMSB, RML, RLL	Chemotherapy, EBRT	5/24/2006 (56)
2-2	7/1/2001 (58)	IIIB sq	RLL, RBI, RMSB	Chemotherapy, PDT + stent	4/24/2002 (59)

Abbreviations: APC, argon plasma coagulation; EBRT, external beam radiotherapy; LLL, left lower lobe; LMSB, left mainstem bronchus; LUL, left upper lobe; mod, moderately differentiated; poor, poorly differentiated; R Hilum, right Hilum; RBI, right bronchus intermedius; RLL, right lower lobe; RMSB, right mainstem bronchus; RUL, right upper lobe; sq, squamous; undif, undifferentiated.

usually becomes clinically evident within 2–24 h. Currently, Photofrin® (dihematoporphyrin ether; Axcan Scandipharm, Birmingham, AL) is the only PS licensed by the FDA for use in advanced-stage lung cancer in the United States. Photofrin® is activated by 630 nm red light, which can be generated by a diode laser and delivered endobronchially via optical fibers fitted with a cylindrical diffusing tip. An endobronchial PDT treatment must be followed in 24 h by a salvage bronchoscopy to clear necrotic tumor slough.

A dose of 2 mg/kg of the FDA-approved PS, Photofrin®, was injected intravenously, after which patients were instructed to avoid exposure to sunlight and bright indoor light for 4–6 weeks. At 48 h post-injection, the patients underwent bronchoscopy during which a fiber-optic device with a diffuser attachment was introduced through a bronchoscope and used to apply 630 nm red light onto the tumor, illuminating it evenly to a dose of 200 J/cm². At 24 h post-treatment, patients underwent salvage bronchoscopy to debride the necrotic tumor and evaluate the response to treatment. If indicated, a second PDT treatment was delivered at this session, followed by a second salvage bronchoscopy 24 h later.

Follow-up

All patients had routine follow-up by bronchoscopy monthly for the first 3 months, then every 3–6 months or as indicated by symptoms. The HDR and PDT procedures were well tolerated and all nine patients completed their therapy regimens, with the exception of patient 1-7, as mentioned above, who had an abbreviated HDR course. No patients were lost to follow-up.

Results

A summary of the results of the HDR and PDT treatments are shown in Table 2. Patient narratives are presented in Table 3.

Discussion

Lung cancer is the most commonly diagnosed cancer and the leading cause of cancer death in the US [13] and worldwide [14], and remains at epidemic levels in large part to widespread tobacco abuse [15]. While most in situ and early-stage (T₁N₀) NSCLC can be resected or ablated with excellent long term outcome [16], some are deemed inoperable and the majority of NSCLC patients are diagnosed with locally advanced or metastatic disease [1]. For patients with recurrent disease, treatment with chemotherapy and/or EBRT offer survival measured in months [2]. As these late stage patients succumb to local and systemic spread, many tumors will grow (or re-grow) locally as obstructive endobronchial lesions which can become hemorrhagic. The quality and duration of life of an incurable lung cancer patient can be improved through maintenance of pulmonary function and avoidance of occlusive morbidities such as hypoxemia and post-obstructive pneumonia. Endobronchial interventions, while they may not change the natural history of advanced-stage lung cancer, offer pallia-

tive relief of local obstruction [4,8]. These techniques may also be used with curative intent in early-stage tumors.

Bronchoscopic interventions include both ablative and non-ablative therapies. The most common and successful non-ablative intervention is stenting, which can be used to reverse or prevent airway collapse and maintain airway patency under external compression, e.g. by a mass lesion. The stenting procedure itself also has inherent morbidity, which can be minimized by operator experience and appropriate patient selection. Many different types of stents can be used to rapidly improve pulmonary function and relieve acute respiratory distress. Unfortunately, because stenting is best suited for relief of external compression [17], results with stenting alone for endobronchial lesions have been relatively short-lived [8]. Most patients will require stent revisions within a few months [17] and can suffer from such complications as tumor growth through or around the stent, granuloma formation, stent migration, infection, and halitosis [18,19]. Ablative therapies cause partial tumor destruction and may be more appropriate monotherapy for endobronchial lesions than stenting. Laser and other immediate-onset modalities have excellent short-term results, with tumor control measured in weeks or months [5,19], but are highly operator dependent. Endobronchial HDR is a well-described procedure with a relatively large literature showing successful palliation. It has a slower onset of action than laser ablation, but shows long-lasting control, measured in months. The major risk of HDR is a misplaced catheter, leading to over dosage of radiation to healthy tissue and subsequent hemorrhage or fistula creation [20]. PDT is another therapy with delayed onset of action that has shown excellent local control of obstructive lesions; it compares favorably to laser ablation as identified in a randomized trial and, like HDR, offers long-lasting local control measured in months [7,21]. Unlike other therapies, it is a two-stage procedure, requiring systemic pretreatment 48 h before bronchoscopic intervention. Major possible side-effects of PDT include tissue edema, transient hemoptysis, necrosis of healthy tissue, and photosensitivity reactions [4].

In the absence of reliable comparative studies, the bronchoscopic intervention a patient receives is usually based on availability of interventional tools and physician training. Generally, palliative interventions are approached sequentially: as each procedure fails, another is performed. This serial technique continues until the risk of harm to the patient appears to outweigh the potential benefit of additional intervention. As previously mentioned, stenting and thermal ablations have a fast onset of action, and are useful for acute relief of obstruction, but they have a tendency to fail in a matter of weeks. HDR and PDT, while not optimal in emergent situations, result in more prolonged local tumor control. By combining PDT with HDR, we hoped to capitalize on the prolonged control each procedure offers in order to increase the overall duration of tumor control and minimize the total number of interventions to the patient. In the process, the potentially synergistic interaction of these two treatments could be explored.

In considering the preferred order of these treatments, a case can be made for using HDR first, based on practical and theoretical advantages. A clinical benefit to starting with HDR is that it can be performed immediately upon diagnosis of endobronchial disease, while in PDT a 48 h delay is

Table 2 Treatment results.

Patient	Treatment dates (spacing) ^a	Location	Treatment-related morbidities	Local control ^b	Death (age)
Treatment group 1 (TG-1, HDR before PDT)					
1-1	10/3/2003–11/19/2003 (33 days)	RUL	Bronchial contraction	5+ years (ongoing)	Alive (73)
1-2	11/1/2006–1/11/2007 (55 days)	RMSB	None	No	7/20/2007 (72)
1-3	1/3/2003–2/5/2003 (20 days)	LMSB	Bronchial contraction	3 months (until death)	5/30/2003 (74)
1-4	8/31/2006–10/4/2006 (20 days)	LUL, Lingula	Complete occlusion from bronchial contraction	2+ years (ongoing)	Alive (66)
1-5	3/22/2004–4/21/2004 (16 days)	RUL/RBI	Complete occlusion from bronchial contraction, Photosensitivity reaction	5+ years (ongoing), with return to baseline PFTs	Alive (69)
1-6	4/15/2003–5/7/2003 (9 days)	RLL	Bronchial contraction	15 months	7/26/2007 (65)
1-7 ^c	12/18/2002–1/7/2003 (20 days)	Distal Trachea	Bronchial contraction, photosensitivity reaction	2 years (until death)	3/2/2005 (55)
Treatment group 2 (TG-2, PDT before HDR)					
2-1	6/19/2006–8/30/2006 (59 days)	LUL, Lingula, LLL	Bronchial contraction	No	9/20/2006 (56)
2-2 ^d	4/30/2002–7/11/2002 (63 days)	RMSB	None	84 days (until 9/24/2002)	7/2/2003 (60)

Abbreviations: LLL, left lower lobe; LMSB, left mainstem bronchus; LUL, left upper lobe; R Hilum, right Hilum; RBI, right bronchus intermedius; RLL, right lower lobe; RMSB, right mainstem bronchus; RUL, right upper lobe.

^a Treatment spacing is measured from completion date of the first treatment modality to the start date of the second.

^b Local control is measured from the completion of both treatments to the date of biopsy-proven recurrence. Current status and ages are as of December 2008.

^c This patient received only 500 cGy of HDR.

^d This patient received only 1000 cGy of HDR.

Table 3 Patient narratives.

Patient	Narrative
1-1	A 64 y/o male with a 50 pack-year smoking Hx, COPD, and CHF presents with a URI. He has a past Hx of 2 foci of poorly diff LUL carcinoma, stage IIIB, 6 years previous s/p LUL lobectomy. A suspicious RUL lesion is found on CXR and needle biopsied, showing poorly diff SCC. He undergoes a wedge resection, but presents 3 years later with hemoptysis and congestion. Bronch shows an occlusive endobronchial tumor in the RUL. He is not a surgical candidate, so he undergoes HDR and PDT therapy to the RUL without complications. 1 and 3-month follow-up bronchs showed tissue growth at the Tx site, but biopsies were negative and the growth was cryo-ablated. Over the following 2 years, there was slow, concentric growth of benign granulation tissue at the treated area of the RUL, which was reversed with electrocautery and balloon dilatation 4 times before it stabilized. Over the next 2 years, the patient developed two new hypermetabolic right lung nodules on PET-CT, for which he has received EBRT. Currently, his RLL nodule is still growing, but there is NED in the RUL more than 5 years after HDR/PDT Tx, at age 73.
1-2	A 69 y/o male with a 55 pack-year smoking Hx, COPD, and PAD with carotid stenosis, presented at an outside hospital with severe dyspnea and hemoptysis. He underwent bronch with Nd-YAG laser resection of an occluding mass in the RUL, which was biopsy positive for SCC. PET-CT showed additional activity in the R Hilum and mediastinal lymph nodes, making his disease stage IV with metastases confined to the lungs. He received palliative EBRT to the affected lung fields to 6140 cGy in a total of 31 fractions with a 3 week break in the middle, resulting in significant tumor shrinkage. He was asymptomatic for over a year, before he experienced a recurrence of the occluding RUL tumor, which was re-treated with laser ablation, followed by HDR/PDT Tx to the RUL, RBI with the use of a SEMS. In the interim between HDR and PDT, tumor had grown through the stent, so an interstitial PDT catheter was used to deliver light internally to the tumor. After 5 days, the bronchus was 100% patent, but had become completely re-occluded within 6 weeks post-Tx, causing severe dyspnea. At 3 months, the occlusion was cleared by APC coagulation, followed by an additional HDR Tx to the RBI. The tumor re-grew, necessitating repeated APC coagulation and re-stenting until the patient finally succumbed to his disease 6 months after HDR/PDT Tx, at age 72.
1-3	A 73 y/o male with a 40 pack-year smoking Hx, COPD, CHF, HTN, DM-2, and CVA presents with weight loss and progressive hoarseness. Imaging shows a small RUL mass as well as a 4 cm × 5 cm mass in the aortic-pulmonary lymph nodes pressing on the recurrent laryngeal nerve. He was staged as IIIB, and biopsy shows SCC. He received carboplatin/Taxol chemotherapy and EBRT at an outside facility, but had to be stented for a 90% occlusion of his LMSB with tumor. He was then given HDR/PDT, followed soon after by gemcitabine chemotherapy and palliative EBRT treatments. His bronchus continued to contract with biopsy negative inflammatory tissue until his death 3 months after his HDR/PDT Tx, at age 74.
1-4	A 64 y/o male with a 50 pack-year smoking history, COPD, and PAD, presents with increasing fatigue and hemoptysis. Imaging shows an active RUL mass and a smaller endobronchial lesion in the left upper lobe. PET-CT and biopsies support a Dx of dual synchronous primary tumors, a poorly diff stage IIB in RUL and a well diff stage IA SCC in the LUL. He immediately began chemotherapy with irinotecan and cisplatin, during which he received EBRT to his RUL lesion and HDR/PDT to his small endobronchial LUL lesion. Between the HDR and PDT Tx, there was repeated formation of obstructive mucus plugs in the LUL. At follow-up bronch one month after PDT, he was without obstructive symptoms, and had NED in the LUL, but had developed dysphagia due to radiation esophagitis from his EBRT Tx. Over the following weeks of chemotherapy and EBRT, he developed inflammatory reactions in both lungs that eventually resolved so that he was able to travel abroad 6 months after his initial Dx. Over the next year, he had several abx-responsive pneumonias. At his 2-year follow-up, he was with NED, but a PET-active LLL lesion was seen on imaging. He is now over 2 years post-HDR/PDT Tx, asymptomatic at age 66 with SBRT planned. He still works.
1-5	A 63 y/o male with a 90 pack-year smoker who presented with a 6 months Hx of hemoptysis. An obstructing mass in the RMSB and Carina was seen on CT, 4 bronchs were performed to take biopsy samples and/or perform APC debulking. He was diagnosed with SCC of RUL, stage IIIB and received 2 cycles of Taxol, platinum, 5-FU and Leucovorin chemotherapy with poor response, followed by HDR and then PDT Tx. Tumor slough from brisk PDT response required early unscheduled debridement. He also had a small pruritic skin rash on his left arm 2 months after Tx that was probably not PDT related. Regular follow-up bronchs showed luminal narrowing due to post-PDT inflammatory changes, improved by cryotherapy, stabilizing after 6 months with no evidence of disease, followed by placement of a SEMS. "Granulation tissue" overgrew the stent, which was removed after 6 weeks. His RMSB continues to experience shrinkage from tissue growth with NED. Now, over 5 years post-HDR/PDT Tx, at age 69, he has bronchs twice a year to open his airway, and is largely asymptomatic between Tx, with baseline PFTs.

Table 3 (Continued)

Patient	Narrative
1-6	A 61 y/o female with a significant smoking history, COPD, asthmatic bronchitis, and obstructive sleep apnea, presents with 1 month of cough productive of blood-tinged sputum. Bronch showed a stage IIA endobronchial tumor in the RBI extending into the RLL. Over the next month, she received HDR/PDT therapy without complications. 3 weeks after Tx, she experienced a RLL pneumonia due to a mucus plug occlusion in the Tx area. Over the following year, she experienced repeated post-obstructive pneumonias associated with benign tissue contraction in the Tx area, reversed at bronch. At 15 months post-Tx, a biopsy of an occluded section revealed recurrent tumor, at which time she underwent EBRT and carboplatin/taxol chemotherapy. The following 3 years were marked by declining lung function, relieved by bronchoscopic interventions. She was admitted repeatedly for pneumonias and recurrent right pleural effusions requiring talc pleurodesis, until her death, 2 years post-HDR/PDT Tx, at age 65.
1-7	A 52 y/o male with a 30 pack-year smoking history, presents with respiratory distress, unresolved with antibiotics. A CT scan showed a 2 cm mass in the distal trachea. Bronchoscopic biopsy found a stage IIIB squamous cell carcinoma of inferior trachea, carina and bilateral proximal mainstem bronchi. He underwent debulking by electrocautery followed by HDR/PDT treatment. At the patient's request, he only received the first of his 3 scheduled HDR doses, preferring EBRT. He received a SEMS and a short course of EBRT before the PDT Tx and the remainder of his EBRT after PDT to a total of 5400 cGy. This EBRT was given at an outside facility and caused significant, but temporary, odynophagia from esophageal exposure. 3 months after his PDT Tx, he complained of "sun sensitivity" which was a painless darkening of exposed areas. He was without respiratory symptoms for 10 months post-Tx, then developed increasing dyspnea and bronch showed narrowing of his trachea and carina with benign tissue, as well as a "floppy distal trachea with dynamic collapse." He received a polyflex stent to the trachea, which migrated and had to be replaced a month later. After an asymptomatic year, a severe pneumonia resulted in intubation. Bronch showed his stent overgrown by benign "granulation tissue." It was removed and PDT Tx delivered. He was soon discharged, but died at home a month later, 2 years post-HDR/PDT Tx, at age 54.
2-1	A 56 y/o male with a 30 pack-year smoking history and COPD, presents for consultation for dyspnea 9 months after a Dx of a stage IIIB poorly diff SCC, s/p EBRT and chemotherapy. Bronch showed endobronchial recurrence in the LUL, LLL, and Lingula. He was scheduled for palliative PDT/HDR Tx. He had 3 SEMS placed, followed by PDT Tx to all 3 areas of recurrence. Tumor slough from brisk PDT response required early unscheduled debridement. Continued declining lung function and stent revisions delayed HDR until 2 months post-PDT. He died with disease 3 weeks after completion of HDR, at age 56.
2-2	A 59 y/o male with a 60 pack-year smoking Hx and COPD presents with post-obstructive right lung collapse 10 months after a diagnosis of stage IIIB SCC of the RLL, s/p PDT Tx, stenting, and one course of Gemzar/carboplatin chemotherapy, completed 5 months prior and with NED in the interim. Bronch showed tumor re-occlusion of the RMSB. He received PDT/HDR treatment with a "fabulous response," pathology negative, seen after only 2 fractions of HDR. The third HDR treatment was not delivered. 12 weeks post-PDT/HDR, he had facial swelling and other signs of SVC compression. Chest CT revealed an enlarging mediastinal mass, and bronch showed new endobronchial tumor growth. Over the next year, he received 4 rounds of chemotherapy, 4 additional 500 cGy fractions of HDR to the RMSB, and a course of EBRT to the RUL and mediastinum. The EBRT resulted in a trachea-esophageal fistula, which was treated with a stent. He died in hospice 1 year post-PDT/HDR Tx, at age 60.

Abbreviations: 5-FU, 5-fluorouracil; abx, antibiotic(s); APC, argon plasma coagulation; bronch, bronchoscopy; CHF, congestive heart failure; COPD, chronic obstructive pulmonary disease; CT, computed tomography; CVA, cerebrovascular accident (stroke); CXR, chest X-ray; diff, differentiated; DM-2, type 2 diabetes mellitus; Dx, diagnosis; EBRT, external beam radiotherapy; HTN, hypertension; Hx, history; LLL, left lower lobe; LMSB, left mainstem bronchus; LUL, left upper lobe; NED, no evidence of disease; PAD, peripheral artery disease; PET, positron emission tomography; PFT, pulmonary function tests; R Hilum, right Hilum; RBI, right bronchus intermedius; RLL, right lower lobe; RMSB, right mainstem bronchus; RUL, right upper lobe; s/p, status post; SBRT, stereotactic body radiation therapy; SCC, squamous cell carcinoma; SEMS, self-expanding metallic stent; SVC, superior vena cava; Sx, symptoms; URI, upper respiratory infection; y/o, year-old.

required for PS accumulation. In addition, the ability of the ionizing radiation in HDR to control active bleeding [22] may justify its use before PDT; much of the red light used to activate the PS in PDT would be absorbed by the blood, limiting tumor exposure. Furthermore, since accurate placement of PDT relies heavily on the operator's ability to visualize all surfaces of the tumor, prior HDR treatment may assist by shrinking the tumor and making its surfaces more clearly

visible. On the other hand, as Freitag et al. point out [9], HDR can reduce vascularization of the tumor over time, which could ultimately inhibit PS uptake and decrease the effectiveness of subsequent PDT treatments. Closer spacing of the treatments in time may circumvent this theoretical effect.

In our study, combined PDT and HDR treatments were well tolerated with all patients but one achieving their pre-

scribed doses. When HDR preceded PDT (TG-1), all but one patient had prolonged clinical and pathological local tumor control (3 months to 5+ years, ongoing), requiring only occasional surveillance bronchoscopy to take biopsy samples and reverse bronchial contraction; none suffered from hemoptysis. Of note, the only patient in TG-1 who failed treatment already had systemic (stage 4) disease. Patients 1-3 and 1-7 died from non-treatment-related causes 3 months and 2 years after treatment, respectively, and were found to be tumor negative on autopsy. Patient 1-6 had recurrence of her cancer after 15 months of local control, and eventually succumbed to her disease. Three of the seven patients in TG-1 are still alive two or more years after their HDR/PDT treatments with ongoing local tumor control, although one of these patients has developed new tumor growth in the opposite lung. The tissue inflammation and contraction seen in many of the patients in TG-1 was bothersome, but easily reversed at bronchoscopy and may have been dose-related. In contrast, the patients in TG-2 had less satisfactory outcomes, with successful local palliation in only one patient, lasting only 10 weeks.

Shorter time between treatments in our study was associated with prolonged local tumor control. Among the nine patients, prolonged local tumor control was achieved in all cases in which the HDR and PDT treatments were given a month or less apart (measured from the completion of one to the start of the other), but in none of the other patients.

While this series is small, it offers evidence that bronchoscopic treatment using sequential HDR and PDT can offer prolonged periods of disease control in patients with endobronchial NSCLC, during which little or no endobronchial intervention is required. This control can perhaps be measured in months and years rather than the weeks and months of other modalities or of HDR and PDT monotherapies. In addition, this study yields some evidence that treatment order and spacing may be crucial variables in determining the efficacy of this approach. Specifically, using HDR before PDT and delivering the treatments no more than a month apart yielded more satisfactory results in this patient population, though it is impossible to know yet which effect plays the larger role.

The excessive soft-tissue responses seen in some patients in this study may have been due to over-dosing of the HDR and/or PDT treatments. While ionizing radiation is often used to prevent tissue overgrowth, it is also known to provoke a local inflammatory response. Prescribing the HDR dose to the luminal surface rather than 5 mm depth may result in less of a tissue reaction. Likewise, the depth of PDT and HDR dose penetration into the tumor may have been uneven. In future series, imaging modalities such as endobronchial ultrasound might aid in gauging the shape and thickness of the tumor more accurately in order to both assess the feasibility of successful treatment and to prescribe a more conformal dose distribution. Lessening the side-effects of the HDR/PDT through refined dosimetry could reduce or eliminate the need for additional endobronchial interventions.

The use of combined HDR/PDT as a bronchoscopic therapy shows great promise for the treatment of endobronchial NSCLC, but more data is needed before this treatment can be strongly recommended. We have yet to fully characterize the effects of treatment order and timing on tumor

response, as well as the relationship between dose and bronchial tissue contraction. Because of the large number of patients who could benefit from this treatment, it behooves us to formulate an HDR/PDT regimen that will maximize tumor control while keeping treatment-related morbidity to a minimum. Small retrospective studies such as ours provide some insight, but a randomized multicenter trial will probably be required in order to clarify these issues.

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